PROJECT REPORT AND HANDOVER

Optimal Sensor Placement for Wastewater Network Monitoring

What was built, how it works, what it found, and how to run, maintain and extend it

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This report records the engineering work completed by the team: a browser-based sensor-placement sandbox built on openly published South Australian sewer data, three EPA SWMM hydraulic simulation studies, and the public site that presents them. It describes the data, models, algorithms, results, verification, assumptions and limitations, and gives a successor everything needed to reproduce, operate and extend the work. It is the companion to the team's compiled research paper, which records the literature and data evidence the work was built on.

Abstract

The project asks where a small number of level sensors should be placed in a gravity sewer so that blockages and overcapacity are seen early. The team built and published a working answer for a real network, the Walkerville council area of metropolitan Adelaide, using only anonymously accessible public data: 1,002 gravity mains with invert levels on every record, 360 maintenance holes, 2,851 connected properties, a 1 m contour layer and the statewide gravity-main layer.

Two complementary tools were produced. The first is a static, browser-only *sandbox* that implements and extends the topology-and-elevation method of Ninh et al. [2025]. It replaces that method's global depth coefficient with an exact flood-fill ceiling over measured cover levels, adds a declared detection threshold and a starved-flow term, and expresses every objective (nodes, pipe length, surcharging chambers, blockage likelihood and growth capacity) as a per-node weight vector consumed by seven built-in algorithms and a user-written one. A Manning screening model supplies capacity, surcharge and growth headroom, and a declared, literature-ordered model supplies blockage likelihood. The second is a series of EPA SWMM dynamic-wave studies. Simulation 1 measures alarm lead time before a spill, Simulation 2 explores where growth causes surcharge, and Simulation 2.5 solves 4,260 growth scenarios on a 71-manhole segment and 8,200 on the whole 328-manhole domain under the quantities that cannot yet be measured, then chooses sensor sets that hold across them.

The principal findings are as follows. Substituting pipe diameter for manhole depth, as the published method does when depth is unavailable, makes 83.2% of chambers appear blind, against 3.2% with reconstructed depth. The published invert fields are labelled by flow direction, not by digitising direction. Greedy coverage outperforms every topology heuristic at every budget, and heuristics anchored on large upstream catchments perform worse than random placement. The single best manhole for detecting growth changes with every unknown, but two or three consensus sensors retain about 92% of each scenario's own optimum under a 10 mm rise rule. No result has been validated against a recorded blockage or overflow, because none are public. The report closes with an operating manual, the extension points, the state of two unmerged branches, the known gaps, and the data the client would need to provide to move from a structural result to an operational recommendation.

Keywords: gravity sewer; optimal sensor placement; observability; maximum coverage; submodular greedy; Manning; EPA SWMM; growth; overcapacity; blockage likelihood; handover.

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1 Project purpose, scope, and deliverables

1.1 The operational problem

Gravity sewers fail in two ways that end the same way, with sewage escaping from a maintenance hole or a property's overflow relief gully. A *blockage* (roots, fat, wipes) stops flow so water backs up behind it. *Overcapacity* needs no obstruction: more flow arrives than a reach was sized for, usually because new housing has connected upstream or infiltration is high in wet weather. Utilities fit level sensors in maintenance holes to see either early, but a network has many more chambers than any budget has sensors. The project brief asks where float and level sensors should go so that a minimum number of devices gives high monitoring capability, with earlier detection of overflow and overcapacity as the intended value [PG S2-22 project team, 2026b,c].

The team's compiled research paper reviewed sensors, placement methods, condition factors and the South Australian data landscape, and recommended a staged, purpose-specific method. This report records what was then *built* to test that direction on a real network, and what it found.

1.2 What was delivered

Table 1: Deliverables in the repository.

Deliverable	What it is	Location
Placement sandbox	Interactive tool. Places a sensor budget on the Walkerville network and compares seven algorithms plus a custom one across five objectives and three observability models, with 2D, 3D and long-section views and a generated assumptions register.	`osp_sandbox.html`, `src/`, `data/`
Explainers	Two standalone teaching pages: four rotatable 3D maintenance-hole cutaways, and a draggable long section showing the backwater wedge and why observability is limited.	`manhole_anatomy.html`, `why_observability.html`
Hydraulic simulations	EPA SWMM models built from the same public data, with the exact input snapshot and every run's output committed for replication.	`simulation_src/`
Simulation pages	Precomputed browser views of Simulation 1 (warning time), 2 (growth, superseded) and 2.5 (growth and sensor placement).	`simulation_legacy/`, `simulation/`, `simulation25/`
Landing site	Tabbed overview hosting every page, with findings, limits and sources, configurable by one file.	`index.html`, `config.js`
Tooling	Node regression suite, data formatter, and a public builder that regenerates the Simulation 2.5 page from committed outputs and checks it byte for byte.	`tools/`
Documentation	README (findings and limits), ARCHITECTURE (module map and rules), simulation README (reproduction), and the in-tool Assumptions, Q&A, Method and Glossary tabs.	repository root, `simulation_src/`

1.3 Scope boundaries

Table 2: Boundaries of the delivered work.

Area	Included	Boundary
Network	Walkerville gravity network (council republication of the operator's register), extended with statewide pipes and pump-station catchments for the simulations.	One council area. The sandbox is clipped to the council boundary, not to a drainage catchment.
Events	Blockage observability (sandbox) and overcapacity from load, growth and infiltration (sandbox screening model and SWMM).	No contaminant, gas or structural-condition monitoring.
Sensors	Level sensors (continuous rise and 150 mm alarm) and float switches (Simulation 1).	No procurement, telemetry, power or installation design.
Data	Only anonymously queryable public services.	No client data, incident history or sensor readings were available.
Validation	Internal consistency, cross-source checks, SWMM mass balance, regression tests.	No validation against a recorded blockage or overflow.

1.4 How to read the numbers in this report

Every figure is computed from the shipped dataset or read from committed SWMM output, and is reproducible with the commands in Section 13. They are *structural* results under declared assumptions: which chambers would see an event under each model, not whether they did. Counts that depend on a declared input, such as demand or infiltration, are always given with that input attached. Figures marked as computed for this report were produced on 13 October 2026 from the repository at commit `8959e2d`.

2 Project approach and timeline

2.1 Approach

The work was organised around three principles that recur through the code and documentation.

1. **Measure before modelling.** Fields were checked for usable values rather than names. Every proxy was replaced by a measurement where one existed, and each replacement was checked to see whether it changed the answer.
2. **Declare what cannot be measured.** Unmeasurable quantities (demand, infiltration, detection threshold, risk weights) are exposed as settings or cases, sourced where possible, and listed in one canonical register generated from the loaded data.
3. **Separate screening from hydraulics.** The browser models are fast, transparent screening calculations. Questions that need backwater, storage and time are answered by EPA SWMM offline, and the browser shows the saved answer.

2.2 Timeline

Table 3: Build timeline from the repository history (62 commits on `main`, 3 September to 5 October 2026).

Dates	Phase	Main changes
2–4 Sep	Sandbox and observability rebuild	Sandbox and explainers from public data. Flood-fill ceiling replaces the published global coefficient, measured depth replaces the diameter proxy, detection threshold and starved-flow term added. 3D relief, tabbed landing page.
10 Sep	Capacity and growth	Manning screening model; the assumptions register becomes canonical.
14 Sep	Attributes and weighting	Per-pipe attributes in the dataset; every algorithm weighted by the objective; blockage-likelihood objective; regression suite; code moved into `src/` and data into `data/`; ARCHITECTURE written.
16–18 Sep	Simulations 1 and 2	SWMM neighbourhood model published; long section with SWMM playback in the sandbox; steady-state growth model with wet-weather and growth-size controls; pull request #1 merged.
22–23 Sep	Branch work	Simulation 2 hydraulic and blockage-timeline extensions on `feature/sim2-hydraulics` (not merged, Section 13.5).
24–25 Sep	Simulation 2.5	Extended domain and nested models, 12-case growth grid, graded detection; growth-headroom objective in the sandbox; SWMM sources and every run committed; per-material Manning's n and a full re-run; reproducible public page builder.
27 Sep–3 Oct	Configuration and whole area	`config.js` page switches; cases defined in one file; placement over all 328 manholes; pipe and manhole detail panels; pump stations drawn separately.
5 Oct	Merge	Pull request #2 merged; `main` at `8959e2d`.
8 Oct	Branch work	Multi-site growth scenarios, a local SWMM runner, and the sandbox methods scored on SWMM, on `wimukthi` (not merged).

2.3 Team practice

Work proceeded in two-week sprints with academic supervision and client meetings [PG S2-22 project team, 2026a,b]. Code was developed on feature branches and merged by pull request. Several engineering rules came out of incidents and are now enforced by tests. A merge once committed unresolved conflict markers and shipped a broken sandbox, so the suite now refuses conflict markers, tracked `.orig` files and unparsable scripts. Editors that format on save reverted the generated data file twice, so data files are generated only. Every declared number now cites its source where it is declared.

3 System architecture

3.1 A static site with offline hydraulics

The site has no server, framework, bundler or tracking. Every file is served as written, so a reviewer can open any file and read the method, and an automated sanitiser can audit the published output. The only external library, three.js, is loaded lazily from a CDN import map when a 3D view is opened. Computation that cannot run in a browser, the SWMM solves, runs offline in Python. Its results ship as quantised, base64-packed JavaScript data files, so the browser shows exact SWMM answers by lookup.

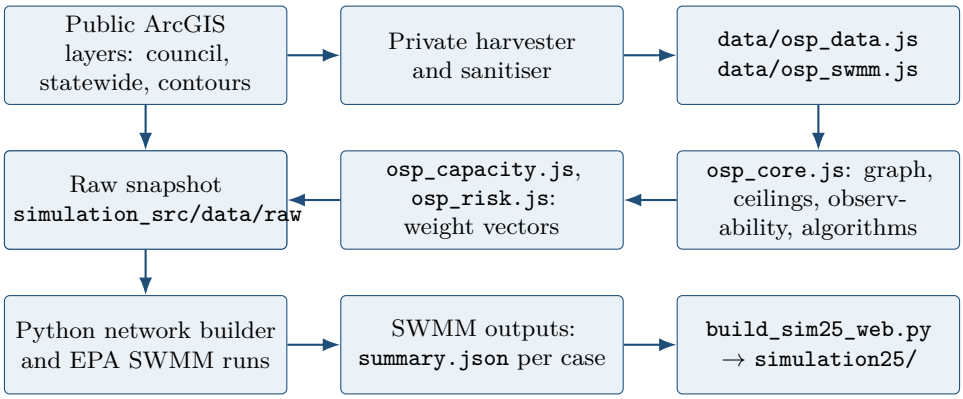


Figure 1: Data flow. Top: the sandbox path. Bottom: the simulation path. The two share the same public sources.

3.2 Repository layout

Table 4: Repository layout and responsibilities.

Path	Responsibility
`index.html`, `config.js`	Landing page with one tab per page (iframes, cache-keyed); `config.js` turns pages on or off.
`osp_sandbox.html`	Sandbox page: loads data, then models, then views, then the UI last.
`manhole_anatomy.html`, `why_observability.html`	Standalone explainers needing no data file.
`src/model/osp_core.js`	Graph in compressed sparse row form, flood-fill ceilings, observability, scoring, every placement algorithm.
`src/model/osp_capacity.js`	Edge geometry, part-full Manning solver, flow accumulation, surcharge, growth, growth headroom and growth cover.
`src/model/osp_risk.js`	Blockage-likelihood factors, blend, chamber aggregation, factor agreement.
`src/ui/osp_ui.js`	All sandbox state, 2D canvas, controls, custom-algorithm Web Worker, anchor mode.
`src/ui/osp_3d.js`, `src/ui/osp_profile.js`	three.js relief view; long-section view (Manning or SWMM water level).
`src/docs/osp_docs.js`	Assumptions register (Parts A–F), Q&A, Method and sources, illustrated glossary, all generated from loaded data.
`data/osp_data.js`, `data/osp_swmm.js`	Generated network dataset and SWMM long-section playback. Never edited by hand.
`simulation_src/`	Python network builder, SWMM models, experiments, raw data snapshot, all outputs, tests, desktop 3D viewer, Windows launcher.
`simulation25/`, `simulation/`, `simulation_legacy/`	Generated browser pages for Simulations 2.5, 2 and 1.
`tools/`	`test_sandbox.js`, `format_data.py`, `build_sim25_web.py`.

3.3 Design rules that hold the system together

1. **Models are pure.** The model files have no DOM access and are wrapped UMD-style, so they load unchanged in the page, in a Web Worker and in Node. This is what allows the regression suite to run without a browser.

2. **Every objective is a per-node weight vector.** `weightOf()` in `osp_core.js` is the only place that knows what an objective means. "Maximise nodes" is every weight set to 1. The capacity, risk and growth models reach the optimiser only by producing a vector, so every algorithm is weightable, and a new objective needs no algorithm change. Weights change what is *worth* seeing. They never change what a sensor *can* see, which depends on geometry alone.
3. **One candidate pool.** Every algorithm draws from `feasible()`: real chambers that observe at least one node. Without this, scores would not be comparable between algorithms.
4. **The Assumptions tab is canonical.** Each assumption is registered once, its figures are computed from the loaded data, and other documents point to it rather than restating it.
5. **Script order is dependency order, and cache keys move together.** Every `?v=N` on pages and scripts is bumped at once.
6. **Data files are generated.** They are written by builders and never edited by hand.
7. **Public output of a private repository.** The harvester (`build_demo_data.py`) and sanitiser (`build_public_demo.py`) live in a private repository. The sanitiser removes private regions and identifying phrases, then audits its own output and refuses to finish if a forbidden term survives. Both scripts are gitignored here; finding either tracked would be a leak.

4 Data and network reconstruction

4.1 Public sources

All requests are anonymous and ask the server for GDA94 / MGA zone 54 (EPSG:28354) coordinates in metres. The layers disagree natively (manholes in wkid 8059, contours in 28354), and a client-side mix-up fails silently [Paul Jaison, 2026a].

Table 5: Public services used.

Source	Content	Use
Council sewer network service	Gravity mains (layer 6), manholes (layer 2), property inspection points (layer 4); attributes NOMINALDIA, INTERNALDI, MATERIAL, MATERIALUN, CONST_YEAR, GRADE, JOINTTYPE.	Network geometry, inverts, chambers, connected properties, pipe attributes.
Council 1 m contours	6,944 contour lines.	Cover-level interpolation.
Statewide gravity mains	20,636 pipes; 77 pumping mains.	Cross-checking the council copy; extending the simulation domain.
Barossa community scheme layer and 5 m contours	Surveyed surface levels.	Validating the contour method only; not shipped publicly.

The simulation reads only a committed raw snapshot (`simulation_src/data/raw/`): 1,002 mains, 360 manholes, 2,851 property inspection points, 2,919 property connections, 533 inspection openings, 15 shafts, 7 rising-main pieces, 5 pumping structures, the contours, and the statewide layers. `fetch_data.py` can re-download them, but the live layers change, so the snapshot is what makes published results reproducible.

4.2 Building the directed graph

Nodes are pipe ends snapped within 1 m. A node is a *chamber*, and so a sensor candidate, only if a published manhole lies within 1 m of it. All 360 manholes matched, with a median snap of 0.38 m and a 90th percentile of 0.55 m, and three duplicates were merged. The sandbox network has 1,010 nodes and 1,001 pipes (one degenerate pipe was skipped), in 14 weakly connected components; the largest has 874 nodes. The assembled graph is acyclic, which was checked rather than assumed. Flow direction came from the publisher's FLOWDIRECT field on 998 pipes and from geometry on three, and no pipe runs uphill after orientation.

4.3 Field semantics: the invert fields are flow-anchored

`START_INVE` is the invert at the *upstream-of-flow* end, whichever way the line was digitised, and FLOWDIRECT (1: first vertex upstream; 2: last) says which geometric end that is. This was established by testing which reading makes inverts agree where two pipes share a chamber:

Table 6: Share of shared chambers with exactly agreeing inverts under each reading.

Reading	Council network	Statewide layer
Geometry-anchored (start = first digitised vertex)	35.2%	32.6%
Orient each pipe by comparing its inverts	37.0%	36.4%
Flow-anchored (start = upstream of flow)	**95.3%**	**85.0%**

The geometry-anchored reading reverses 583 of 1,002 mains while leaving every record, node and edge count unchanged, so it survives every obvious sanity check. Read correctly, invert inconsistencies at chambers fall from 462 to 4. The common check that `START_INVE > END_INVERT` holds on every record is therefore true by definition of the field names. It is not evidence that the network runs downhill or is acyclic.

4.4 Pipe attributes

Attributes were re-queried from the same mains layer and joined back by geometry rather than by identifier, so the join can be re-checked.

Table 7: Attribute completeness on the 1,001 modelled reaches and how each is used.

Attribute	Published	Use
Nominal diameter	100%	Capacity, likelihood, map line width. Equals the earlier $\min(d_u, d_v)$ proxy on all 1,001 reaches.
Material	100%	Manning's n by material; likelihood. MATERIAL reads UNKN on 456 mains, but MATERIALUN reads VC on all 456 and agrees on all 43 records carrying both. Result: 906 vitrified clay, 94 uPVC, 1 reinforced concrete.
Construction year	99.5%	Likelihood (age). Range 1896–2017, median 1912; 91.7% over fifty years old.
Grade	99.9%	Rescues flat or adverse reaches in the capacity model; likelihood. Median ratio to fall over length 1.0000, with 796 reaches within ±5%, which independently confirms the flow-anchored reading.
Internal diameter	54.4%	Deliberately unused: it covers only the newer half, and since $Q \propto D^{8/3}$ mixing bases would bias the ranking.
Joint type	54.4%	Likelihood only, at low weight.
Roughness	0%	Unusable.

4.5 Chamber depth reconstruction

Depth (cover minus invert) is almost never published. Cover level was interpolated from the 1 m contours and validated three ways:

Table 8: Cover-level validation.

Comparison	n	Mean abs. error	p90	RMSE
1 m contours vs surveyed covers, same area	10	**0.23 m**	0.41 m	0.29 m
5 m contours vs surveyed levels, external network	707	1.34 m	2.62 m	2.08 m
5 m subset vs full 1 m layer, same terrain	1,009	1.85 m	4.54 m	2.59 m

The second and third rows measure the same error by two routes and agree, which is what makes the first credible on a small sample. The external set excluded 354 points outside contour coverage and 693 failing the publisher's own surface-minus-invert identity. The resulting depths have a 10th percentile of 1.87 m, a median of 2.92 m and a 90th percentile of 3.92 m (10 surveyed, 999 interpolated, 1 discarded with cover below invert).

4.6 Council and statewide overlap, and the simulation domain

`overlap_check.py` compares every council main against its statewide copy before any statewide pipe is used. 999 of 1,002 match on asset identifier. Diameters and both inverts agree on all 999, flow direction agrees on 989, and geometry agrees on 966 (median offset 4 mm; 33 drawn differently, one trunk by about 48 m). A datum shift of (0.648, 1.498) m is applied to the statewide frame. The check also lists every boundary pipe as an inflow, outflow or *gap* (a council branch joined to the rest only through a link the council does not hold).

For the simulations, statewide-only pipes are joined to the council network by asset identifier and flow-anchored end, not by coordinate snapping, which misses real joins. Outside nodes are sealed junctions and never sensor candidates. Outside load is estimated from pipe length at the connection density measured on council pipes of the same size class; trunks above 450 mm receive none. Rising-main pieces are chained end to end before matching, giving three pump stations, two of which drain into the domain.

Connected properties are counted per pipe from the inspection points (matched to a lateral within 15 m, else to the nearest pipe within 40 m). Above the segment outlet this gives 643 properties on 7,781 m of sewer, 8.3 per 100 m, which replaced an earlier assumed 13 per 100 m that overstated population by about 1.6 times.

5 Blockage observability and placement (the sandbox)

5.1 Problem formulation

Let $G = (V, E)$ be the directed acyclic network oriented by flow, $C \subseteq V$ the chambers, and $O(s) \subseteq V$ the nodes at which a blockage would be detected by a sensor at s. For a budget k and weights w, placement is the weighted maximum-coverage problem

$$\max_{S \subseteq C,\ |S| \le k} \quad f(S) = \sum_{v \in V} w_v \, \mathbf{1}(\exists s \in S : v \in O(s)) \, .$$

The problem is NP-hard, but f is monotone submodular, so greedy selection is within a factor $1 - 1/e$ of optimal [Nemhauser et al., 1978]. The contribution of the sandbox is in how $O(s)$ is computed and in what w can represent.

5.2 The ceiling: where backed-up water escapes

When node v blocks, flow stops and the water surface behind it is horizontal. It rises until it reaches the lowest opening in the flooded region, a manhole cover or a property's overflow relief gully (ORG). The ORG is designed by AS/NZS 3500.2 to be the lowest opening, but it is published nowhere, so the sandbox offers a proxy 0.1 m above ground [Standards Australia, 2021]. The escape level is called $\mathrm{ceil}(v)$. Raising a level L, the flooded set $R(L)$ grows monotonically while $\min\{\mathrm{opening}(u) : u \in R(L)\}$ cannot increase, so there is a single crossing point. `ceilings()` finds it exactly with a min-heap keyed on invert. Starting at v, it pops nodes in increasing invert order, updates the running minimum opening c, pushes each popped node's *upstream* neighbours, and stops as soon as the next invert is at or above c.

5.3 Observability models

For each candidate s, a depth-first traversal *downstream* tests each reached v:

Table 9: Selectable observability models.

Model	Condition	Purpose
Measured headroom (default)	$\mathrm{ceil}(v) - \mathrm{inv}(s) > \tau$	Measured geometry and a declared detection threshold τ (default 0.05 m).
As published	$\mathrm{inv}(v) + c \cdot d(v) > \mathrm{inv}(s)$, $c = 0.7$	Ninh et al. [2025] with pipe diameter d standing in for depth, kept to show the cost of the proxy.
Elevation budget	$\mathrm{inv}(s) - \mathrm{inv}(v) < D$	Plain fall budget (default 1 m).

Because the backed-up surface is flat, $\mathrm{ceil}(v) - \mathrm{inv}(s)$ is the water depth at s with no gradient term. The published reach formula $(c \times \mathrm{MaxDepth})/\mathrm{gradient}$ is the constant-gradient special case. Three deliberate departures from the published method were made on 2 September 2026: the global coefficient was removed, because it is a per-node physical quantity and raising it claims headroom the network does not have; measured depth replaced the diameter proxy; and a declared threshold was added, because a 2 mm rise should not count the same as 2 m. An optional *starved-flow* term also credits a sensor downstream of a blockage when the stopped branch is at least a set share of the flow that normally passes it. Pruning the traversal by elevation was tried and rejected, because node elevation does not fall monotonically downstream and pruning silently dropped observable nodes.

5.4 Objectives as weight vectors

Table 10: Objectives offered and the weight each assigns to observing node v.

Objective	**Weight** w_v
Maximise nodes observed	1
Maximise pipe length observed	Length of pipe flowing into v
Maximise surcharging chambers observed	1 where the capacity model surcharges v, else 0
Maximise blockage risk observed	Likelihood exposure of v (Section 6.2)
Maximise watched growth capacity	Number of candidate connection sites whose first failure surcharges v (Section 6.1)

5.5 Placement algorithms

Table 11: Algorithms in the sandbox. All score through `weightOf` and draw from the same feasible pool.

Algorithm	**Function**	**Method**
Greedy set cover (CELF)	`greedy`	Repeatedly adds the sensor with the largest marginal weighted gain. Lazy evaluation re-checks only the top of a gain-sorted queue, which is exact because coverage is submodular [Leskovec et al., 2007]. With an unlimited budget it gives the full-cover set. The benchmark.
Two up, two down	`twoUpTwoDown`	Field rule of thumb. Anchors in order of upstream weight, groups kept from overlapping, then k chambers up and k down breadth-first. Only the choice of anchor follows the objective; the rule itself is not weighted.
Largest upstream catchment	`upstreamSize`	Weight strictly upstream, in one topological pass.
Highest betweenness	`betweenness`	Brandes betweenness, vertex-weighted by source [Brandes, 2001].
Highest out-degree / in-degree	`degreeWeight`	Weighted degree; out-degree reported as the better simple heuristic by Wang et al. [2024].
Random (best of 20)	`randomPlace`	Best of 20 draws from the feasible pool, which makes it a stronger baseline than a uniform draw.
Custom JavaScript	Web Worker	The user writes `place(graph, budget, api)`; the API offers candidates, observable sets, objective weights, incoming pipes with attributes, and downstream traversal.

Anchor mode prices a mandated location. The user picks a chamber, and `greedyForced` reports the coverage with that chamber forced into the set against the unconstrained optimum. Nodes without a manhole are refused.

6 Capacity, growth, and blockage-likelihood models

6.1 Capacity screening

`osp_capacity.js` answers a different question from blockage: where accumulated flow exceeds what a reach can carry. It is a screening calculation (steady, uniform, normal depth, with no backwater, routing, storage or time), and the code and the tool label it so. Where it disagrees with SWMM, SWMM is right.

Edge geometry. Diameter is the published nominal diameter. Slope is fall over length between flow-anchored inverts. Where that is flat or adverse, the publisher's GRADE is used (three reaches rescued), and only when neither is usable is the slope clamped to 10^{-4} (no reach clamped). Manning's n comes from material: 0.013 for vitrified clay and reinforced concrete, 0.010 for uPVC, and 0.013 as the fallback [Chow, 1959]. As Alshami et al. [2023] note, roughness in practice is usually assumed rather than measured.

Part-full circular section. With wetted angle θ:

$$A = \tfrac{D^2}{8}(\theta - \sin\theta), \quad P = \tfrac{D\theta}{2}, \quad R = \tfrac{A}{P}, \quad Q = \tfrac{1}{n} A R^{2/3} S^{1/2}, \quad \tfrac{d}{D} = \tfrac{1-\cos(\theta/2)}{2}.$$

Flow peaks at $\theta \approx 5.2781$ ($d/D \approx 0.938$), slightly above full-bore flow, so the greatest passable flow $q_{\max}$ and the full-bore flow are kept separate.

Accumulation and surcharge. A per-node load (default 0.05 L/s) is pushed down the graph in topological order, split equally between outgoing pipes, and multiplied by a peak factor. A reach whose flow exceeds $q_{\max}$ is over capacity, and the chamber *above* it surcharges. The surcharging set feeds the observability machinery unchanged, because once a chamber surcharges the cause no longer matters to whether a sensor sees it.

Growth headroom. For each node v, one unit of load is injected and pushed downstream with the same split rule. With spare capacity $\text{spare}(e) = q_{\max}(e) - q(e)$ and $f(v \to e)$ the fraction of v's load that reaches e,

$$\text{headroom}(v) = \min_{e \text{ downstream of } v} \frac{\text{spare}(e)}{f(v \to e)}.$$

This gives the new load v can take before something tips, and the *binding reach*. Superposition makes it exact. The regression suite confirms that adding 99% of the headroom at the tightest site tips nothing and adding 101% tips exactly one reach. `growthCover` turns the result into an objective: each chamber is weighted by the number of connection sites whose first failure surcharges it. The weight is a count with no severity term, by design.

6.2 Blockage likelihood

Ninh et al. [2025] names risk weighting as the obvious extension of its method, and Crowley et al. [2025] names equal node weighting as a limitation. `osp_risk.js` joins that gap to the condition literature. Each reach is scored on factors normalised to $[0, 1]$, where 1 means more likely to block, and unpublished values score a neutral 0.5:

Table 12: Blockage-likelihood factors and default weights.

Factor	Weight	Scoring and source of direction
Age	0.30	Oldest in network scores 1. Ranked first by mutual information (0.124) in Ma et al. [2025].
Diameter	0.25	Smallest bore scores 1. Ranked second (0.032) in Ma et al. [2025]; small-bore reticulation carries most blockages [Drenoyanis and Prackwieser, 2022]; Malek Mohammadi et al. [2020] report contradictory evidence, hence a moderate weight.
Gradient	0.20	$1 - g/1\%$, zero at or above a self-cleansing 1%; flat slopes cause deposition and septicity [Malek Mohammadi et al., 2020].
Material	0.15	VC 1.0, RC 0.7, uPVC 0.3, ranked on root intrusion at joints [Drenoyanis and Prackwieser, 2022]. Values are declared.
Joint type	0.05	Bitumen 1.0, spigot and socket 0.8, plastic 0.4, rubber ring 0.3; published on only 54%.
Length	0.05	Saturating at 100 m.

The order follows the only measured ranking in the literature reviewed. The spacing (1.2:1 between age and diameter rather than about 4:1) is a declared choice. Imported thresholds should not be trusted [Malek Mohammadi et al., 2020, Ma et al., 2025], and on this network age and bore rank-correlate at -0.32, because the 1896 sewers are the large trunks and the small-bore reticulation came later. Reach scores aggregate to chambers by *exposure* (length-weighted sum of incoming reaches) or *intensity* (how bad a chamber's pipes are). Measured on the shipped data, reach scores span 0.316 to 0.882. Rank agreement of each single factor with the blend is: age 0.78, material 0.50, gradient 0.28, joint 0.23, length 0.07 and diameter -0.13. No single factor reproduces the blend, so the weighting is the method rather than a detail of it, and the defensible claim is the ranking of reaches, not the score.

7 Sandbox results

7.1 The depth proxy, not the terrain, made the network look blind

Table 13: Observability model comparison on the Walkerville network (1,010 nodes; feasible pool 356 chambers). Computed for this report.

Model	Observable universe	Chambers observing nothing	Nodes covered by greedy, 40 sensors	Sensors for full cover
As published ($c = 0.7$, diameter)	157	840 (83.2%)	80	60
Elevation budget, 1 m	673	230	207	201
Measured headroom, $\tau = 0.05$ m	**813**	**32 (3.2%)**	**341**	**168**
Measured headroom, $\tau = 0.15$ m	811	35	336	167
Headroom with starved flow at 0.5	946	5	432	189

Pipe diameter (0.15 m) understates the depth term by roughly nineteen times. With measured depth, nearly every chamber can see something. The answer is almost insensitive to the threshold between 50 and 150 mm, because the headroom available is metres, not millimetres.

7.2 Algorithm comparison

Table 14: Share of the observable objective covered by each algorithm (headroom model, $\tau = 0.05$ m; random is best of 20 seeded draws, with the mean in brackets). Computed for this report.

Objective	k	Greedy	2 up 2 down	Upstream	Betw.	Out-deg.	In-deg.	Random
Nodes	10	19.9	2.5	2.5	6.0	9.0	9.5	12.5 (9.3)
Nodes	40	41.9	8.6	8.6	9.3	22.6	19.2	27.7 (23.7)
Pipe length	40	49.6	11.8	11.9	12.9	26.8	21.2	33.9 (28.4)
Surcharging (0.4 L/s)	5	91.3	8.8	8.8	46.3	50.0	77.5	66.3 (37.5)
Surcharging (0.4 L/s)	10	98.8	22.5	22.5	51.3	67.5	77.5	67.5 (50.7)
Risk exposure	40	50.2	12.0	11.9	13.0	26.4	22.2	34.0 (28.5)
Growth capacity	10	45.9	2.8	2.8	5.4	34.4	20.0	27.6 (15.8)
Growth capacity	40	77.5	19.4	20.8	19.2	56.1	27.9	49.6 (41.5)

Four observations follow.

1. Greedy dominates at every budget and objective.
2. Heuristics anchored on large upstream catchments (two up two down, largest catchment) perform *worse than random*. Under this model a sensor sees blockages downstream of it, because water backs up to it, so anchors chosen near outlets have little below them. Out-degree is the best simple heuristic, consistent with Wang et al. [2024].
3. Overcapacity is concentrated. At 0.4 L/s per node, 80 chambers surcharge, and five greedy sensors observe 91% of them, ten observe 99%. A placement optimised for blockage is not optimised for overcapacity, which is why the objectives are kept separate.
4. Full blockage coverage needs 168 sensors, against 360 chambers.

7.3 Capacity and growth

Surcharge counts depend on the declared demand. At 0.05, 0.1, 0.2, 0.4 and 0.9 L/s per node (peak factor 1), 0, 3, 30, 80 and 132 chambers surcharge. A surcharge count is therefore quoted only with its demand case. At the

default load, 992 nodes have a finite growth headroom and none is under 1 L/s. The 359 candidate connection sites resolve onto only 133 binding chambers, which is the structure set cover exploits.

8 Hydraulic simulations with EPA SWMM

8.1 Engine and common settings

All three studies use EPA SWMM 5 through PySWMM [Rossman, 2015, McDonnell et al., 2020], Python 3.11, with pinned versions (`pyswmm` 2.1.0, `swmm-toolkit` 0.17.0, NumPy 2.3.3, SciPy 1.17.1).

Table 15: SWMM options used throughout.

Option	Value and reason
Routing	Dynamic wave (full Saint-Venant equations), so backwater, storage and surcharge are real. Routing step 0.5 s, report step 10 s, inertial damping partial, normal-flow limited both.
Surcharge method	Preissmann SLOT, not EXTRAN. On a test ramp, EXTRAN junctions stored no water while surcharged (the shaft rose 1.3 m on 0 m^3) and EXTRAN storage nodes behaved as 3.54 m^2 with 12.4% continuity error; the slot gave 0.93 m^2 against a true 0.87 m^2 with 0.1% error.
Chambers	1,050 mm plan diameter; in-pipe segment nodes a token 0.02 m^2.
Roughness	Manning's n by material as in the sandbox. Only Simulation 2.5 has been re-run with it; Simulations 1 and 2 used a uniform 0.013.
Stages	Half pipe, *surcharge* (above the outlet crown, what a level sensor sees), half shaft, spill.
Outputs	Every run keeps `model.inp`, `.rpt` and `.out`; Simulation 1 runs also keep `series.npz` and `meta.json`.

8.2 Simulation 1: warning time

A four-chamber neighbourhood (A, B, J, D) at junction 441 was used to ask how long before the first spill a level sensor or a float switch would alarm. Load was applied three ways: equally per chamber, by upstream sewer length lumped at chambers, or *along* each pipe in roughly 15 m segments, as house connections deliver it. Scenarios were constant, ramp, step and event (60 min warm-up, 30 min rise, 30 min hold, 60 min fall).

Table 16: Load ladder (along split, L/s per 100 m of sewer): steady load at which each chamber first reaches each stage. 61 runs, bisection to 2%, worst continuity error −0.38%.

Chamber	Half pipe	Surcharge	Half shaft	Spill
A	0.167	0.368	0.828	1.22
B	0.828	0.855	1.12	22.6
J	0.164	0.38	0.975	never
D	0.16	0.404	never	never

The first surcharge anywhere occurs at 4.9 times indicative dry-weather flow, and the first spill at 16.2 times.

Table 17: Warning lead time in minutes before the first spill (at A). Level sensor alarms at +0.10 m over its settled base; float switch trips at the outlet crown.

Event peak	First spill (min)	A level / float	B	J	D
1.25 × spill load	25.7	20.5 / 20.0	7.7 / 7.3	19.5 / 19.0	18.0 / 17.5
2 ×	16.3	13.0 / 12.7	4.3 / 4.2	12.2 / 11.8	11.0 / 10.7
3 ×	11.3	9.0 / 8.8	2.5 / 2.5	8.2 / 8.0	7.2 / 7.0

A level sensor gains only 0.3 to 0.5 minutes over a float at the same chamber, while the choice of chamber changes the lead time by up to 13 minutes. Location matters far more than device. The growth experiment showed that A, B, J and D can take 17 to 20 L/s of extra constant load before a new chamber surcharges. The split comparison showed that the loading method changes *where* stages appear first (D under equal loading, A under length-based loading). This page is hidden by `config.js` because Simulation 2.5 supersedes its question for placement.

8.3 Simulation 2: growth (superseded)

The whole catchment above node 583 was modelled in steady state: 157 pipes, 71 chambers and 643 connected dwellings at 500 L per dwelling per day with peak factor 2. This retired the earlier assumption that the 7.8 km above A arrives as an instantaneous lump. Two measurements shaped it. Growth alone cannot surcharge this catchment: at peak dry-weather flow the worst reach runs at 28% of full bore and the median at 0.8%, so every run carries infiltration. And the catchment is complete in the published data, with 36 of 38 headwaters inside the layer.

The first pass added 500 dwellings at every site, saturating the answer. Median site capacity was 140 dwellings, so two chambers tipped in 69 of 71 scenarios and greedy returned a single sensor. Re-running each site at its own capacity gave two sensors covering all 71. The growth grid (infiltration 0.25, 0.4 and 0.55 L/s per 100 m by growth of 50, 150, 350 and 700 dwellings) gave the page its controls. Simulation 2 was superseded because three outside pipes (about 3.2 km) also drain into the catchment, and because a yes/no threshold made the answer hinge on whichever manhole happened to sit just below it.

8.4 Simulation 2.5: growth and sensor placement under the unknowns

Model. Two models are built from the same network and used together (*nesting*). The *whole* model is every council pipe draining to node 606, where the council data ends, plus the pump-station catchments and exit pipes, with a free or fixed-level outfall. The *segment* model is the 71 manholes above node 583, with its outlet pipe discharging at the level the whole model settled to. Where an outside pipe flows in, a constant inflow sized from statewide pipe length stands in for it. Because SWMM accepts one inlet per outfall, multi-pipe outfall nodes are converted to junctions and connected to the outfall by a 1 m dummy pipe with a 1 mm fall. Runs settle for 480 minutes, within 0.2 mm at every weather level tested.

Loads. Each connection carries 2.81 people (8,023 people over 2,851 connections, ABS 2021), at 200 L per person per day. A new dwelling carries 2.3 people. The peak factor is 2.0, or the Harmon formula $1 + 14/(4 + \sqrt{P/1000})$ [Harmon, 1918]. The overnight trough is 0.2 times average flow. Infiltration and inflow are given in L/s per 100 m of pipe, optionally weighted by age band (1.5, 1.2, 0.8 and 0.5 for pipes built by 1929, 1959, 1989 and later).

Cases: what cannot be measured yet. Cases are defined only in `cases.json`, which is validated on import. Weather levels are 0.0 (dry), 0.11 (design wet, which just meets the utility's DN225 design rating), 0.25 (beyond design) and 0.40 (severe, 2.4 times the design envelope). Sensitivity variants override one field each: Harmon peaking, age-weighted infiltration, outside inflow $\times 0.5$ or $\times 1.5$, a fixed outfall stage, or the trough hour. The committed segment grid has 12 cases (`cases_all.json`). The current `cases.json` runs the whole area with dry, design wet and severe plus two variants at design wet, five cases in all.

Growth grid. Every study manhole receives 10, 30, 80, 150 or (stress) 700 new dwellings. On the segment that is 355 runs per case and 4,260 across 12 cases. Nesting was validated at design wet (identical detections in 12 of 12 cases, depth difference 0.0 mm) and used up to that level at about 3.5 s per run. It *failed* at 0.25 (0 of 12), because a surcharged pipe discharging into a fixed-level outfall settles about 10 mm higher than into the real manhole, so wetter cases run on the whole domain. The whole-area grid places growth at any of 328 manholes, 1,641 runs per case at about 30 to 35 s each. The five-case grid took 8 h 22 min. Rises below 1 mm are not stored, which keeps a case at 0.8 to 2 MB instead of about 13 MB. A corridor family spreads 80 to 485 dwellings along Walkerville Terrace; even in severe weather it raises no study manhole by more than 0.8 mm, because none of the terrace lies in the study segment.

Detection rules and placement. A manhole *sees* a scenario under one of the following rules:

- **alarm**: its depth newly passes the operator's 150 mm low alarm [Do et al., 2023];
- **surcharge**: it newly surcharges;
- **graded**: its depth rises at least 10, 25 or 50 mm over its own baseline.

The graded rules were added because, with an absolute threshold, one marginal manhole detects nearly everything, and which manhole that is moves with every unknown. 50 mm is roughly the width of the 20 to 75 mm normal daily range reported for a real deployment [Do et al., 2023]. Greedy set cover over scenarios gives each case's own set. Robustness is how often each manhole is chosen across cases. For the page, one nested 1 to 10 sensor order is chosen across all cases, either for the best *average* coverage or for the best *worst case* (max-min).

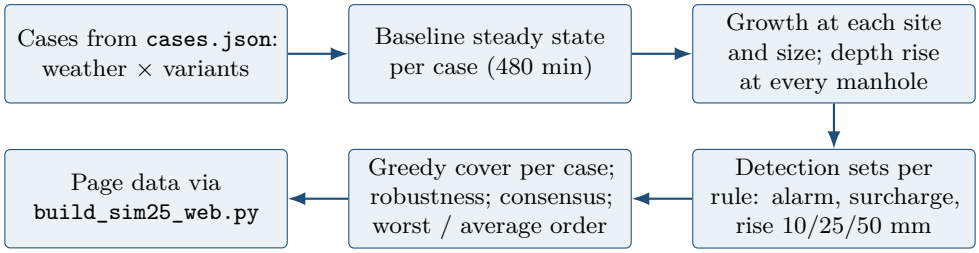


Figure 2: Simulation 2.5 pipeline.

Table 18: Per-case results on the segment. Baseline counts are manholes over the alarm, then surcharged, before growth; detectable is out of 355 scenarios.

Case	Method	Baseline	Greedy alarm set	Detectable
Dry	nested	0 / 0	4449882	1
Design wet	nested	2 / 0	4448419, 4449103	127
Design wet, inflow ×0.5	nested	0 / 0	4450193	282
Design wet, inflow ×1.5	nested	7 / 6	4447394, 4448853	162
Design wet, age-weighted	nested	2 / 0	4448419, 4449173	85
Design wet, Harmon	nested	9 / 9	4429499	190
Beyond design	whole	33 / 33	4428836	349
Beyond design, inflow ×0.5	whole	22 / 22	4447128	278
Beyond design, inflow ×1.5	whole	41 / 41	4428765	331
Beyond design, age-weighted	whole	31 / 31	4447880	255
Beyond design, Harmon	whole	38 / 38	4428765, 4449503, 4429645, 4450100	204
Severe	whole	45 / 45	4428268	104

Results on the 71-manhole segment (12 cases). Under the absolute alarm rule no manhole is chosen in more than 2 of the 12 cases. The best site changes with every unknown. The graded rules restore a stable answer:

Table 19: Consensus placement under graded rules: one set chosen across all 12 cases, and the share of each case's own best coverage it keeps (range, mean).

Rule	1 sensor	2 sensors	3 sensors
Rise 10 mm	4448419: 51–100%, mean 85%	+ 4450193: 75–100%, mean 92%	+ 4429312: 78–100%, mean 93%
Rise 25 mm	4449785: 18–97%, mean 82%	+ 4447394: 84–100%, mean 92%	+ 4450332: 84–100%, mean 92%
Rise 50 mm	mean 72% (0–100%)	mean 75%	mean 81%

At 50 mm, so few scenarios are visible in the dry and design-wet cases (40 to 69 of 355) that no set is robust.

Table 20: Whole-area results; detectable is out of 1,640 growth scenarios.

Case	Baseline	Greedy alarm set	Detectable
Dry	21 / 0	4428096, 4451823, 4449882, 4453770	11
Design wet	34 / 7	4450531, 4448419, 4453770	368
Design wet, age-weighted	34 / 7	4450531, 4448419, 4428096, 4451823, 4453770	222
Design wet, inflow ×1.5	53 / 29	4448853, 4453489, 4450137, 4427588, 4447394, 4453770	453
Severe	147 / 141	4428268, 4452946, 4451287, 1623826, 4449436, 4453770	249

Results on the whole 328-manhole area (five cases, current configuration). Over the whole area one manhole, **4453770**, is chosen in all five cases under the alarm rule and in four of five under the surcharge rule. The next most robust (4428096, 4451823, 4450531) appear in two of five. Most growth scenarios are not detectable by any manhole under the alarm rule (1,272 of 1,640 at design wet): small developments far from a bottleneck move no level by the alarm margin.

The daily cycle. The peak and trough of the day were each solved as steady states. In dry weather the modelled peak level at a typical segment manhole is 13.5 mm (median; 10th to 90th percentile 6.5 to 55.7 mm), and only 28% of manholes fall within the 20 to 75 mm range reported for the operator's deployment [Do et al., 2023]. At design wet the figures are 19.9 mm and 37%. Modelled small-pipe levels therefore sit at or below the low end of the observed range. For a sample of 150-dwelling developments, the rise at the peak hour (11.7 to 17.3 mm) is six to seven times the rise at the trough (1.6 to 2.9 mm), so growth is best seen at the peak of the day. The true detection threshold cannot be set from the model, because it depends on day-to-day variation at a given hour, which only real level records contain.

9 Web application and user interface

9.1 Pages

Table 21: Pages on the site.

Tab	Content	Shown
Overview	Findings, honest limits, data sources, links into each page.	always
What a maintenance hole is	Four rotatable 3D sections: one chamber inside, why every chamber on a run is a different depth, a chamber taking several pipes, and where a blockage sends the water.	yes
Why observability is limited	Draggable long section of the backwater wedge and the arithmetic that limits how far a sensor can see; why a steep catchment looks blind.	yes
OSP Sandbox	The tool (Section 5).	always
Simulation 1	Warning time: one scenario, three event sizes, alarm markers, a phase banner explaining the 60-minute warm-up.	no
Simulation 2	Growth: homes behind a manhole, bottleneck pipes, Q&A. Superseded.	no
Simulation 2.5	Growth and sensor placement (below).	yes

9.2 The sandbox interface

The sidebar holds the following controls:

- region, and mode (network or anchor);
- observability model with its threshold, ORG toggle, coefficient c, fall budget and starved-flow share;
- sensor budget, objective and algorithm (with k up and k down, or custom code);
- colour-by: coverage, invert, depth, capacity d/D, diameter, likelihood or growth headroom;
- likelihood blend and aggregation;
- capacity load, peak factor and Manning's n (by material or fixed);
- growth additions;
- 3D exaggeration and water display;
- long-section span, exaggeration and water source.

The centre shows a 2D canvas, the three.js relief, or the long section. The long section draws water level from either the Manning model, which responds to the sliders but cannot show backwater, or precomputed SWMM playback of three scenarios (dry, high load, and wet weather with infiltration). It always states which is on screen, because the difference between the two pictures is the argument for SWMM. The right panel reports the result and keeps a deduplicated comparison list. Further tabs hold the Assumptions register, Q&A, Method and sources, and an illustrated glossary.

9.3 The Simulation 2.5 page

The user picks where new dwellings connect by clicking a manhole, then sets the growth size, case, detection rule, sensor count (1 to 10) and objective (worst case or average). The map colours manholes grey (room to spare), amber (already surcharged before growth) or red (tipped by this growth). Amber and red are never merged, because a reach that was already a problem is not evidence about growth. Optional layers show a heatmap ranking every manhole by its own coverage, homes and their laterals, pump stations, a pipe details card, and a side panel for the selected manhole. Elevation is exaggerated, with the factor shown on screen.

10 Verification and quality assurance

Table 22: Verification in place and its state at handover.

Check	What it covers	State, 13 Oct 2026
`nodetools/test_sandbox.js`	63 checks. Repository hygiene (conflict markers, merge leftovers, every script parses, dependency order, data parses, configuration wiring); 3D meshes coloured before compile; graph counts; pinned observability, capacity, headroom and likelihood figures; every heuristic reproduces its unweighted form with all-ones weights and responds to risk; weighting never changes observability; a region without attributes degrades gracefully; documentation renders with no undefined or NaN.	63 passed, 0 failed
`pythonsimulation_src/test_simulation.py`	SWMM invariants: pipe counts, load totals, outfall flow equals inflow, continuity under 2% (1% whole domain), flow accumulation, shaft storage as a 1,050 mm chamber, scenario shapes, rising-main chaining, 71 study manholes, three outside inflows, nesting within 1 mm at design wet.	Passing at last commit; not re-run here (dependencies not installed)
`pythontools/build_sim25_web.py--check`	Rebuilds the Simulation 2.5 page from committed outputs and compares every generated file byte for byte.	Not re-run here
`pythontools/format_data.py--check`	Data file formatting stable.	Not re-run here
SWMM mass balance	Worst continuity error recorded per experiment (−0.38% ladder, −0.26% growth).	Recorded in outputs
Cross-source checks	Flow-anchored inverts, GRADE agreement, overlap with the statewide copy, depth validation.	Section 4

The tests are regression checks on one dataset, not unit tests of the algorithms. If the data changes, the expected figures must be re-established deliberately.

11 Consolidated findings

1. **A depth proxy, not the terrain, was making the network look unobservable.** With pipe diameter in place of depth, 840 of 1,010 chambers (83.2%) observe nothing; with reconstructed depth, 32 (3.2%).
2. **The invert fields are flow-anchored.** Read correctly, invert agreement at chambers is 95.3%; read by geometry, 583 of 1,002 mains are silently reversed.
3. **Fields that look empty are not the only field.** MATERIAL is unknown on 456 mains; MATERIALUN completes it to 100%. The publisher's GRADE independently confirms the invert reading (median ratio 1.0000).
4. **Two retired assumptions changed nothing, and are reported because they were checked.** Per-pipe diameter equals the old proxy on all 1,001 reaches. The roughness field is populated nowhere.
5. **Optimisation beats rules of thumb.** Greedy coverage leads at every budget and objective. Upstream-anchored rules do worse than random under the blockage model. Unmerged branch work reports the same for growth when every method is scored on SWMM (Section 13.5).
6. **Blockage and overcapacity need different placements.** Overcapacity concentrates at a few bottlenecks (five sensors see 91% of 80 surcharging chambers); blockage coverage needs 168 sensors for full cover.
7. **Location matters more than device.** In Simulation 1, a level sensor gains under half a minute over a float at the same chamber; choosing the chamber changes lead time by up to 13 minutes.
8. **The best single manhole for growth moves with every unknown.** A consensus of two or three sensors retains about 92% of each case's own optimum under a 10 or 25 mm rise rule. Over the whole area, manhole 4453770 is chosen in every case.
9. **Absolute alarm thresholds make placement unstable; graded rises make it stable.** Growth is six to seven times more visible at the daily peak than at the trough.

12 Assumptions and limitations

The Assumptions tab in the sandbox is the canonical register (Part A network and blockage model; B capacity and growth; C what remains assumed; D blockage likelihood; E dynamic model and long section; F growth headroom). The limitations that matter most are stated here.

1. **No validation against a recorded blockage or overflow.** Every result is structural. Incident history is the single highest-value addition.
2. **Detection thresholds are declared.** No South Australian level traces are public, and day-to-day variation at a given hour is needed to set them.
3. **Demand and infiltration are settings.** The surcharge count moves from 0 to 132 across the demand range, so counts are quoted with their case.
4. **Boundaries.** The sandbox is clipped to the council area, so flow from outside is not counted and edge reaches look healthier than they are (the unsafe direction). Simulation 2.5 adds estimated boundary inflows to correct this.
5. **ORG levels are unpublished.** The proxy is reasoned from AS/NZS 3500.2, not measured.
6. **Pumps are not represented in the sandbox**, which treats the network as gravity only. Whether pump stations are in scope changes the method rather than extending it.
7. **The real network deviates from the record.** Ground movement, roots and infiltration are not modelled. A perturbation study (jittering inverts and reporting how far the recommendation moves) is planned but not done.
8. **Likelihood weights are declared, not fitted.** The ranking is the claim, not the score.
9. **The Manning model is screening only.** It has no backwater, which SWMM shows is where most 25 mm rises come from.
10. **Roughness provenance.** Only Simulation 2.5 uses per-material n, and its `summary.json` files record no solver or roughness provenance.

Routes probed and rejected, so that they are not retried:

- national elevation services return HTTP 403;
- the statewide ground-elevation raster is tile-cached only and refuses per-point sampling (HTTP 500);
- no statewide manhole layer exists;
- the statewide roughness field is populated on 0.8% of records.

13 Handover: operating, reproducing, and extending the work

13.1 Getting started

1. Clone the repository and serve it over HTTP: `python -m http.server 8001`, then open http://localhost:8001. Opening files directly breaks the Web Worker and the 3D views.
2. Run `node tools/test_sandbox.js`. It should report 63 passed.
3. For the simulations, install Python 3.11, then run `pip install -r simulation_src/requirements.txt`. On Windows, `simulation_src/run_simulation.bat` installs missing packages, fetches data once, and opens a menu.

13.2 Command reference

Table 23: Commands. Simulation commands run from `simulation_src/` unless stated.

Command	Purpose
`node tools/test_sandbox.js` (root)	Regression and hygiene suite; run before every push and every merge.
`python tools/format_data.py -check` (root)	Would the data file change if reformatted?
`python fetch_data.py [-force]`	Re-download public layers (live data; not an exact reproduction).

Command	Purpose
`python overlap_check.py`	Council versus statewide comparison and boundary list.
`python experiments.py all`	Simulation 1: ladder, growth, warning, compare.
`python hydraulics.py -list`; `python view3d.py <run>`	Single runs; desktop 3D playback (`-screenshot`, `-gif`).
`python growth_grid.py [-sites 8]`	Simulation 2 grid (superseded).
`python sim25.py -describe`	Simulation 2.5 domain, inflows, pumps, loads.
`python sim25_grid.py -plan`	List cases and run counts without running.
`python sim25_grid.py -cases all -workers 6 [-resume]`	Simulation 2.5 grid; `-resume` skips solved cases.
`python sim25_graded.py > results/sim25/graded_report.txt`	Graded-rule placement and consensus.
`python sim25_daily.py`	Daily-cycle validation and per-manhole thresholds (about 2.5 min).
`python test_simulation.py`	Simulation test suite.
`python tools/build_sim25_web.py -check` (root)	Rebuild the page from outputs into `reproduced_simulation25/` and compare byte for byte.
`python tools/build_sim25_web.py -publish` (root)	Validate cases, rebuild, copy into `simulation25/`, bump cache keys. Refuses unsolved cases.

13.3 Maintenance rules

- Bump every `?v=N` cache key together when a page or script changes. At handover the pages are at v=31, Simulation 2.5 at v=48, and `config.js` at v=32.
- Keep script order in `osp_sandbox.html` as data, models, views, then `osp_ui.js`.
- Never edit `data/*.js` or generated simulation data by hand. Close them in editors that format on save.
- Add or remove an assumption in code and in the register in the same commit.
- Merges touching `osp_ui.js` or `osp_docs.js` need care: both sides of a past merge each added a "Part D" with colliding row identifiers.
- Keep `simulation_src/data/raw/` for replication; a fresh fetch substitutes today's data.
- Never commit the private harvester or sanitiser. The repository is public.

13.4 Extension points

Table 24: How to extend the system safely.

Change	Where	Rule
New objective	Produce a per-node weight vector and pass it as `{w}`; add the option in `osp_sandbox.html` and the case in `osp_ui.js`.	No algorithm changes; register the assumption.
New algorithm	`osp_core.js`: score through `weightOf`, draw from `feasible()`; add it to the UI switch.	Add a test that all-ones weights reproduce the unweighted form.
New Simulation 2.5 case	Edit `cases.json` only, run `sim25_grid.py -resume`, then `build_sim25_web.py -publish`.	Validated on import; the page shows only solved cases.
Segment or whole area	`study_area` in `cases.json`.	Results go to separate folders (`grid/`, `grid_whole/`).
Show or hide a page	`config.js`.	Boolean flags only (tested).
New region or client data	Private harvester and sanitiser regenerate `data/osp_data.js`.	Re-establish the pinned test figures deliberately.

13.5 Unmerged work on remote branches

Two branches on the remote hold work that is not on `main` at `8959e2d`. A successor should review and merge or retire them.

Table 25: Unmerged branches (fetched 13 October 2026).

Branch	Content and status
`wimukthi` (2 commits, 8 Oct; 14 files, +1,792/−39)	**Multi-site growth scenarios with a local SWMM runner.** The Simulation 2.5 page can build a scenario across many manholes. An in-page estimate (exact steady flows, Manning, and a hydraulic-grade-line backwater pass) was compared with 8,200 grid runs. It matched SWMM to the millimetre where pipes flow freely and found 99 to 100% of 25 mm rises in dry and design-wet cases, at up to 40% false flags, but only 22% in severe weather. A local runner (`simulation_src/sim25_serve.py`, listening on 127.0.0.1 only) solves custom scenarios with the grid's own function and reproduced a stored run within 0.05 mm. Results are badged SWMM or ESTIMATE. A pump recirculation loop was found by mass balance and handled. **Placement methods scored on SWMM.** The page offers the sandbox's algorithms (vendored `osp_core.js` with a byte-for-byte check) next to SWMM greedy, all scored on the SWMM growth scenarios. At three sensors under a 25 mm rise (worst case / average), the branch reports: SWMM greedy 39% / 60%; flood-fill CELF 4% / 47%; random 7% / 25%; betweenness 0% / 14%; two up two down and largest catchment 0% / 0%. Placement for growth needs the hydraulics. The branch's suite has 69 checks and Simulation 2.5 cache keys at v=50.
`feature/sim2-hydraulics` (27 commits, 22–23 Sep; 17 files, +31,751/−724)	Extends the superseded Simulation 2 page: flow animation, pump stations, a real-time blockage timeline in a pure module (`simulation/model/blockage_physics.js`) with declared choke exponent and shaft diameter, SWMM cross-validation (`tools/validate_blockage_timeline.py`), heatmap and ranking pins, a revised sub-window layout, WebGL power savings, keyboard navigation, `tools/serve.js`, a 730-line test file, and CHANGELOG, DATA_PROVENANCE, FEATURE_DOCUMENTATION and VALIDATION documents. Branched before Simulation 2.5, so conflicts with `simulation/` and `index.html` are expected. Simulation 2 is hidden on `main`.

13.6 Known gaps and stale documentation

1. `ARCHITECTURE.md` says the suite has 38 checks; it has 63.
2. `README.md` says "seven algorithms against four objectives"; there are seven built-in algorithms plus a custom one, and five objectives.
3. Code comments refer to documents held only in the private repository: the assumptions spreadsheet (register identifiers such as L4, P02, X6), the Simulation 2.5 research, method and results notes, ASSUMPTIONS, HANDOFF and HOW_IT_WAS_BUILT. A successor needs access to them, or the identifiers should be resolved into this report's Appendix B.
4. Simulation 2.5 outputs record no provenance (roughness, PySWMM version, routing error); Simulation 1 outputs do. Writing that metadata is a small fix.
5. `simulation_src/README.md` says the packed page data still holds uniform-n answers. The page data was regenerated on 2 October, after the per-material re-run, so the statement is probably stale and should be verified with `-check`.
6. The committed 12-case segment results correspond to `cases_all.json`. `cases.json` currently runs the five-case whole area.
7. Regenerating `data/osp_data.js` or `data/osp_swmm.js` needs the private pipeline (harvester, sanitiser, `build_swmm.py`).

13.7 Recommended next work

1. Review and merge the `wimukthi` branch. It turns the Simulation 2.5 page into a direct, SWMM-scored comparison of the methods and allows custom scenarios.
2. Decide the fate of `feature/sim2-hydraulics`: port the blockage-timeline physics and validation into the current structure, or retire it.

3. Run the planned perturbation study (invert jitter, threshold and demand sweeps) and report placement stability alongside optimality.
4. Add provenance to Simulation 2.5 outputs and re-run Simulations 1 and 2 with per-material roughness if they are to be quoted.
5. With client data: validate against incident history, calibrate thresholds from level traces, replace declared demand and infiltration, and price sensors and access.

13.8 Information needed from the client

These requests follow from the register and match the team's data request plan [Paul Jaison, 2026b]:

- dated choke, overflow and complaint records with locations;
- level traces from deployed sensors, for threshold calibration and day-to-day variation;
- a catchment-complete network with a maintenance-hole layer;
- overflow relief gully levels, or a sample of them;
- dry-weather flow per connection or billed consumption;
- infiltration estimates;
- the operator's hydraulic model;
- a decision on whether pump stations are in scope;
- sensor cost, access, power and telemetry constraints.

14 Conclusion

The project delivered a working, public, reproducible answer to where level sensors should go in a real gravity sewer. It used only open data, kept every unmeasurable quantity visible as a declared setting or case, and tested its screening models against a full hydraulic solver.

Three results carry beyond this network. First, reconstructing depth and reading the invert fields by flow direction, rather than relying on convenient proxies, decides whether a network looks observable at all. Second, optimised placement consistently beats rules of thumb, and blockage and overcapacity call for different placements, so a single generic coverage figure is not a sufficient design target. Third, the best single sensor for detecting growth is not stable across the quantities that cannot yet be measured, but a small consensus set under a graded-rise rule is. That is the defensible form of a recommendation until the client's incident records and level traces allow it to be calibrated.

The system is structured for handover. Pure models are covered by a regression suite, every objective is a weight vector, simulation cases are defined in a single file, the browser page is rebuilt byte for byte from committed outputs, and every assumption is held in a canonical register. The next steps are to merge the outstanding branch work, run the perturbation study, and apply the same pipeline to client data.

A Glossary

Table 26: Terms used in this report.

Term	Meaning in this project
Chamber	A node carrying a published maintenance hole; the only valid sensor location.
Invert	Level of the inside floor of a pipe at an end or chamber.
Cover level	Lid level; surveyed on 10 chambers, interpolated from 1 m contours elsewhere.
Depth	Cover minus invert; the headroom water can climb before escaping at the lid.
Crown	Top inside of a pipe. A chamber surcharges when water is above its outlet crown.
ORG	Overflow relief gully: the property-level opening designed to be the lowest escape point.
Ceiling	Level at which water backed up behind a blockage escapes; found by flood fill.
Observable set	Nodes at which a blockage would raise the level at a sensor by more than the threshold.
Starved flow	Drop in flow at a sensor downstream of a blockage.

Term	Meaning in this project
Flow-anchored	Field convention in which "start" means the upstream-of-flow end.
Headroom (growth)	New load a site can take before the first downstream reach exceeds capacity.
Binding reach	The reach that fails first for a given site.
Tipped	Fine before growth and surcharged after; distinct from already surcharged.
$q_{\max}$	Greatest flow a circular pipe can pass, at $d/D \approx 0.938$.
I&I	Infiltration and inflow, in L/s per 100 m of pipe.
Peak factor	Ratio of peak to average sewage flow; flat 2.0 or Harmon.
Nesting	Solving a segment model with boundary levels handed down from the whole-domain model.
Graded detection	A manhole sees a scenario when its depth rises by a set amount over its own baseline.
Consensus set	One sensor set chosen across all cases, judged by the share of each case's own best that it keeps.
CELF	Cost-effective lazy forward selection: greedy with lazy re-evaluation of marginal gains.
CSR	Compressed sparse row adjacency used for the graph and observable sets.
UMD	Module pattern that loads as a browser global or a Node module.

B Declared constants and register identifiers

Table 27: Constants declared in the simulation code, with the register identifiers used in comments.

ID	Value	Meaning and source
P01	2.81 people per connection	8,023 people over 2,851 connections (ABS 2021).
P01b	200 L/person/day	Sewage generation.
G4	2.3 people per new dwelling	ABS 2021 average household size.
P02	Peak factor 2.0; Harmon high case	Dry-weather to peak.
R9	Trough 0.2 × average	Midpoint of the 0.10 to 0.30 minimum-hour range.
P04	I&I 0.0, 0.11, 0.25, 0.40 L/s per 100 m	0.11 meets the DN225 design envelope.
P26	Age bands 1.5 / 1.2 / 0.8 / 0.5	Built by 1929, 1959, 1989, later; sensitivity only.
P25	Outside-inflow factor 1.0 (0.5, 1.5)	Boundary inflow sensitivity.
P31 / X6	Outfall free, or fixed stage	Downstream boundary at node 606.
G6	150 mm	Operator's low alarm [Do et al., 2023].
P05	10, 30, 80, 150, 700 dwellings; corridor 80, 150, 390, 485	Growth sizes; 390 and 485 are projected 2021–41 medium and high growth.
X1, X2	Node 606; node 583	Domain end; study-segment outlet (71 manholes).
X4	Nesting up to I&I 0.11	Validated to 0.0 mm at design wet; failed at 0.25.
R2	480 minutes	Settling time (within 0.2 mm).
H3	$n = 0.013$ fallback	With 0.010 for uPVC [Chow, 1959].
H8, H10	1,050 mm chamber; 0.02 m^2 inline nodes	SWMM node areas.
H11	SLOT surcharge method	See the SWMM settings table.
E1–E6	120 min; 2%; 0.15 L/s per 100 m base; +0.10 m level alarm; peaks 1.25, 2, 3	Simulation 1 experiment settings.
Sandbox	τ 0.05 m; c 0.7; D 1.0 m; ORG +0.1 m; load 0.05 L/s per node; MIN_SLOPE 10^{-4}	Defaults in `osp_core.js` and `osp_capacity.js`.

C Research notes behind this report

The report was compiled in phases. The intermediate research notes are kept beside it in `reports/` so that each figure can be traced: `research_01_template.md` (format), `research_02_history.md` (timeline and contributors), `research_03_data.md` (data and pipeline), `research_04_sandbox.md` (models and computed comparisons), `research_05_simulations.md` (SWMM studies and results), and `research_06_handover.md` (operations, branches and gaps). `Summary.MD` is a shorter orientation to the repository.

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